GENERATED STARTER SCAFFOLD ONLY — HUMAN ELECTRICAL REVIEW REQUIRED; NOT ELECTRICALLY VERIFIED OR APPROVED FOR FABRICATION

Generated notes only: no symbols, nets, footprints, or pin mappings are approved.

BLE antenna: TODO: VERIFY exact ESP32-S3 module edge placement and datasheet keepout.

NFC antenna: TODO: VERIFY loop geometry, metal exclusions, tuning network, and assembled measurement.

SECTION: Power input and LiPo charger

Candidate direction: USB-C 5 V input, protected LiPo, BQ24074-class power path

Review scope: Input limits, charge current, NTC strategy, status, and thermal path

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: 3.3 V power rail

Candidate direction: TPS63031-class buck-boost with inductor and decoupling

Review scope: Peak load, stability, efficiency, brownout, and power sequencing

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: ESP32-S3 module

Candidate direction: ESP32-S3-MINI-1-N4R2 direction from V1 BOM shortlist

Review scope: Power, EN/BOOT, straps, GPIO allocation, USB, and decoupling

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: USB-C / USB data / ESD

Candidate direction: USB4105-class receptacle and low-capacitance USB ESD

Review scope: CC pull-downs, D+/D-, shield, return path, and connector mechanics

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: 240x320 TFT display connector

Candidate direction: 14-pin 0.5 mm FPC connector for ST7789-class SPI panel

Review scope: FPC contact side, SPI, reset/DC/CS, backlight, and insertion direction

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: Dynamic NFC tag and I2C

Candidate direction: ST25DV04KC-class dynamic tag, I2C pull-ups, and tuning placeholders

Review scope: I2C address, GPO/FD, energy harvesting decision, loop, and measurement pads

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: Optional SPI NOR flash

Candidate direction: W25Q128JV-class 3.3 V SPI NOR; DNP until architecture review

Review scope: Bus sharing, chip select, hold/write protect, decoupling, and DNP safety

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: RGB LEDs

Candidate direction: SK6805-EC15-class side emitters with current budget controls

Review scope: Supply, 3.3 V data margin, direction, decoupling, and optical orientation

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: Buttons

Candidate direction: Three SKRPACE010-class tactile switches

Review scope: GPIO straps, pull-ups, debounce, wake behavior, and actuator direction

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: Piezo buzzer

Candidate direction: PKLCS1212E4001-R1-class passive piezo

Review scope: Drive topology, voltage, resonant frequency, acoustic port, and quiet state

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: Optional vibration motor driver

Candidate direction: DNP motor pads and AO3400A-class low-side switch

Review scope: Stall current, 3.3 V gate drive, flyback, EMI, timeout, and DNP safety

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

SECTION: Test pads

Candidate direction: Bare ENIG pogo pads; no unverified SMT test-point footprints

Review scope: GND, VBUS, VBAT, 3V3, EN, BOOT, UART, USB, I2C, SPI, and switched loads

TODO: VERIFY exact footprint and land pattern before symbol placement.

TODO: VERIFY datasheet-confirmed pinout and electrical limits.

TODO: VERIFY JLCPCB assembly status, orientation, and DNP/manual-install state.

No electrical verification or fabrication approval
Starter/scaffold requiring human electrical review
MCard-StarterKit clean-room hardware study

Sheet: /
File: MCard_V1.kicad_sch

Title: MCard V1 generated schematic scaffold

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 1/1